zotregistry.dev

None

None

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1. What's New

1.1

1.1.1 Workload Identity and Sync Authentication

- OIDC workload identity tokens can now complete the registry token-service login flow, including Kubernetes `imagePullSecret` authentication.
- Sync adds OAuth2 JWT assertion and RFC 8693 token-exchange credential helpers for short-lived upstream credentials.
- A new `gcp` credential helper uses Google Application Default Credentials to mirror from Google Artifact Registry and compatible registries.

See Authentication and authorization and OCI registry mirroring.

1.1.2 Retention and Garbage Collection Controls

- Retention policies can protect actively used untagged manifests, such as digest-only pull-through cache entries, with `keepUntagged`.
- Periodic garbage collection can be restricted to a daily UTC `gcTimeWindow`, including windows that cross midnight.

See Tag retention policies and Storage.

1.1.3 Signing, Scanning, Events, and GraphQL

- Generated notation trust policies now support the `tsa:default` trust store for timestamped signatures.
- Trivy scanning now accepts OCI zstd-compressed layers.
- Successful, non-cached image scans emit `zotregistry.image.scanned` events with vulnerability summaries.
- GraphQL `ImageSummary` now exposes `ArtifactType`.

See Verifying image signatures, CVE scanning, Events, and GraphQL.

1.1.4 Reliability, Conformance, and Security Fixes

- Hardened the OIDC UI callback redirect against open redirects that used a backslash in the target URL.
- Sync registries can set `maxRetryDelay` for exponential HTTP retry backoff (when greater than `retryDelay`).
- Improved OCI conformance for malformed manifest references, referrers responses, and blob digest errors.
- Additional fixes cover digest multi-tag overwrite authorization, cosign verification binding, garbage collection of incomplete content, storage integrity checks, sync diagnostics and credential refresh, Windows repository paths, and metadata robustness.

See OCI registry mirroring and Events.

1.2

1.2.1 Trivy-Based SBOM Artifact Generation

zot now supports generating SBOM artifacts during Trivy CVE scans and publishing them as OCI referrers for scanned images.

For configuration and usage details, see CVE scanning.

1.2.2 Storage: Azure Backend, Fast Restart, and Blob Redirects

This release adds multiple storage improvements:

- Added Azure Blob Storage support as a remote `storageDriver` backend (`name: "azure"`)
- Added optional `storage.fastRestart` to reduce startup time by skipping storage walk when safe
- Added optional `storage.redirectBlobURL` for redirecting blob pulls to backend signed URLs

For details and examples, see Storage.

1.2.3 Authentication and Authorization Enhancements

- Added authorization support for GitHub team memberships in OpenID login flows
- Added metrics endpoint anonymous access controls with stricter authz behavior for authenticated users not in metrics ACL
- Added metrics path validation hardening for safer configuration

See Authentication and authorization and Monitoring.

1.2.4 Event Payload Metadata Improvements

The events extension now includes actor and request metadata in webhook payloads when request context is available.

See Events.

1.2.5 zli Configuration UX

`zli` now supports default named configurations so commands can run without explicitly passing `--config` or `--url` when a default is set.

See zli user guide.

1.2.6 Bug fixes

Many bug fixes around storage behavior, authentication and authorization flows, metrics, CVE reference links, and dependency updates.

1.3

1.3.1 OIDC Logout and Identity Mapping

zot now supports OpenID Connect RP-Initiated Logout and includes improvements to identity handling and claim-mapping logs.

```
{
  "issuer": "https://iam.example.com",
  "scopes": ["openid", "profile", "email", "groups"]
}
```

Authentication mapping also adds support for OpenID groups claims, making it easier to align external identity providers with zot access policies.

```
{
  "claimMapping": {
    "username": "preferred_username",
    "groups": "groups"
  }
}
```

1.3.2 Conditional Authorization via CEL

Authorization policies can now use conditional expressions through CEL (Common Expression Language), enabling more context-aware access control decisions.

```
{
  "conditions": [
    {
      "expression": "req.time < timestamp(\"2099-12-31T23:59:59Z\")",
      "message": "alice's prod access expires end of 2099"
    }
  ]
}
```

1.3.3 Blob Transfer and Upload Handling Improvements

This release includes multiple API and distribution improvements:

- Added support for multipart range blob pulls and multipart download enhancements
- Improved upload range handling (including proper `416` responses for invalid ranges)
- Recognized Docker Compose/Buildx user agents in the v2 auth challenge workaround

1.3.4 HTTP Read/Write Timeout Configuration

HTTP read and write timeouts are now configurable for both the API server and metrics exporter, with defaults set to `60s`.

Use larger values for environments that handle large images or slower client/network paths. You can also set the timeout values to `0` to disable timeouts (infinite timeout) when required.

For configuration examples and behavior details, see Security Posture: HTTP read/write timeout configuration.

1.3.5 zli and Configuration UX Enhancements

Several usability updates were added to `zli`:

- Improved CVE diff output
- Added a typed `~/.zot` config layer with strict validation
- Added config management commands: `list`, `show`, `get`, `set`, and `reset`

1.3.6 Observability, Signing, and UI Updates

- Added Prometheus metrics for garbage collection
- Added support for cosign bundle metadata
- Updated UI dependencies (`zui`) and related integration pieces

1.3.7 Bug fixes

Many bug fixes around authentication, sync filtering, transfer reliability, lint/CI, dependency updates, and other miscellaneous improvements.

1.4

1.4.1 Repository Quota Enforcement

zot now includes repository quota enforcement middleware in the API layer, helping limit repository growth according to configured quota policies.

Configuration summary (from PR #3923): set `storage.maxRepos` to cap how many repositories can be created. When the limit is reached, pushes that create a new repository are rejected (HTTP 429), while pushes to existing repositories continue to work. Setting `maxRepos` to `0` (or omitting it) disables enforcement.

```
{
  "storage": {
    "maxRepos": 10
  }
}
```

1.4.2 Multi-tag Push Support

zot now supports pushing multiple tags for a single manifest, improving tag management workflows and reducing duplicate push operations.

`PUT /v2/<name>/manifests/<digest>?tag=1.2.3&tag=1.2&tag=1&tag=latest`

1.4.3 Configuration Schema Export

A new schema command (`zot schema`) can now dump the JSON Schema for zot configuration, making validation and tooling integration easier.

1.4.4 Storage and Authentication Fixes

- Fixed GCS storage handling for root directory prefixing and EOF behavior during walk operations
- Added workaround for Docker client authentication in mixed anonymous-policy setups
- Prevented repository update flows from corrupting `index.json` when disk space is exhausted

1.4.5 Security Hardening

Several hardening fixes were included in this release:

- Limited manifest PUT request bodies to 4 MiB
- Limited API key creation request bodies to 4 KiB
- Suppressed `Allow-Credentials` when CORS origin is wildcard
- Removed `InsecureSkipVerify` usage from metrics client TLS

1.4.6 UI and Search Metadata Enhancements

- Added `LastPullTimestamp` and `PushedBy` fields to search/index `ImageSummary`
- Updated UI dependencies (`zui`) and related integration improvements

1.4.7 Bug fixes

Many bug fixes around security, storage, CI workflows, and other miscellaneous features.

1.5

1.5.1 GCS Storage Support

zot now supports Google Cloud Storage (GCS) as a storage backend, expanding the available storage options alongside existing S3-compatible storage solutions.

1.5.2 Dynamic TLS Certificate Reloading

TLS certificates can now be automatically reloaded when changed on disk, eliminating the need to restart zot when certificates are renewed. This is particularly useful in environments with short-lived certificates or automated certificate rotation.

1.5.3 Enhanced OIDC and JWT Authentication

Multiple improvements to authentication capabilities:

- **Per-issuer CA support**: Configure custom certificate authorities for individual OIDC issuers
- **AWS Secrets Manager integration**: Support for AWS Secrets Manager for JWT verification key storage
- **Fine-grained JWT expiration**: Configure `exp` claim at the access entry level for more granular token lifetime control

1.5.4 UI and GraphQL Enhancements

- Added `TaggedTimestamp` to `ImageSummary` returned by GraphQL API
- "Last Tagged" timestamp now displayed in the tag details view in zot's UI

1.5.5 Sync Improvements

- **Legacy Cosign tag filtering**: New `SyncLegacyCosignTags` configuration option to selectively skip syncing legacy cosign/SBOM tags
- **Smart OCI conversion**: Skip OCI conversion for already-synced images, improving sync performance

1.5.6 Security Fixes

- Fixed open redirect vulnerability via `callback_ui`
- Ensured "latest" tag authz checks are not skipped during updates

1.5.7 Bug fixes

Many bug fixes around metadata timestamps, Windows path handling, and other miscellaneous features.

1.6

1.6.1 Workload Identity Federation

zot can now authenticate workloads using OIDC ID tokens, enabling secret-less authentication for automated workflows.

See examples/config-bearer-oidc-workload.json for an example configuration.

1.6.2 Bug fixes

Many bug fixes around storage and other miscellaneous features.

1.7

1.7.1 Bug fixes

Fixes a bug where requests having an Authorization header are rejected if basic auth is disabled.

1.8

1.8.1 mTLS Configuration

Previously, there were some access control/authz limitations when enabling UI and mTLS for machine clients. Now mTLS authn configuration supports more flexible identity extraction from peer certs so that access control is uniformly enforced.

1.8.2 Bug fixes

Many bug fixes around security, sync and other miscellaneous features.

1.9

1.9.1 Further FIPS 140-3 Improvements

Previously, password-based authentication supported only *bcrypt* (based on Blowfish) which is not FIPS 140 approved. Starting with this release, both SHA256 and SHA512 based hashes are supported. You can use `mkpasswd` utility (available on most Linux distributions) for this purpose.

1.9.2 Test Retention Policies

A `verify-feature` retention subcommand has been added that allows users to preview and validate retention policy changes without running the actual zot server. The command runs GC and retention tasks in dry-run mode for immediate feedback.

```
zot verify-feature retention -l /var/log/zot-retention-check.log -i 1m -t 10m <config-file>
```

1.9.3 Improved Self-Hosted OAuth2 Support

Some self-hosted applications such as GitHub Enterprise require custom auth url, token url and username claim mapping.

```
"auth": {
    "openid": {
      "providers": {
        "github": {
          "clientid": <client_id>,
          "clientsecret": <client_secret>,
          "authurl": "https://github.company.com/login/oauth/authorize",     // Custom GHE authorization endpoint
          "tokenurl": "https://github.company.com/login/oauth/access_token", // Custom GHE token endpoint
          "scopes": ["read:org", "user", "repo"],
          "claimMapping": {
            "username": "preferred_username"                                 // Custom claim mapping
          }
        }
      }
    }
```

1.9.4 UI Improvements

zot's UI has been migrated to use vite which helps with startup times and long-term project maintenance.

1.9.5 Bug fixes

Many bug fixes around security, sync and other miscellaneous features.

1.10

1.10.1 zot as a Golang Library

zot was originally intended as an application although implemented in golang. Starting from this release zot can also be used as a golang library.

```
import zotregistry.dev/zot/v2
```

However, note that this aspect will be best-effort only.

1.10.2 Bug fixes

Some minor bug fixes.

1.11

1.11.1 Garbage Collection and Retention

Untagged manifests are now removed by default in garbage collection unless they are referenced by indexes or artifacts. Previously they were only removed if the retention policies settings were present in the zot configuration. If unspecified, the configuration option for retention delay (used for untagged manifests and orphan referrers), now defaults to the GC delay instead of 24 hours.

1.11.2 FIPS 140-3 Compliance

On Linux, zot can be deployed in FIPS 140-3 mode (in terms of cryptographic protocols used) by setting the environment variable `GODEBUG="fips140=only"`.

Note that GODEBUG=fips140=on and only are not supported on OpenBSD, Wasm, AIX, and 32-bit Windows platforms.

1.11.3 Bug fixes

Update the GraphQL playground version to 5.2, as the old one had broken online dependencies.

1.12

1.12.1 Bug fixes

Some bug fixes in sync mirroring and garbage collection.

1.13

1.13.1 zot OCI Container Images for FreeBSD

FreeBSD community has been releasing OCI images to enable their container ecosystem.

zot is now releasing OCI images to deploy and run as native FreeBSD containers. For example, `ghcr.io/project-zot/zot-freebsd-amd64:latest`.

1.13.2 Bug fixes

Some minor bug fixes.

1.14

1.14.1 Liveness and Readiness Endpoints for Kubernetes Deployments

New HTTP endpoints are added for improved Kubernetes deployments - `/livez`, `/readyz` and `/startupz`. Helm chart has been updated.

1.14.2 OpenID Credentials Stored in Files

Previously OpenID credentials were specified inline in zot configuration. Now, they can be loaded from a separate file and the file can be constructed as a Kubernetes `Secret`.

1.14.3 Bug fixes

Some bug fixes around FreeBSD builds.

1.15

1.15.1 Gitlab Social Login

Gitlab login is now supported in zot UI.

1.15.2 Events Extension Token and Custom HTTP Header Support

Earlier, events extension for HTTP sink supported only basic authentication. Now token authentication is supported. Furthermore, you can also include customer HTTP headers to enable sink side routing, etc.

1.16

1.16.1 Bug fixes

Some bug fixes around authentication and authorization.

1.17

1.17.1 Event generation

zot can now generate registry-significant events that can be published to http or nats endpoints.

1.17.2 Improved `docker` Support in UI

Native docker images are now correctly displayed in the Web UI.

1.17.3 Redis driver Support

Previous releases supported a local BoltDB or a remote DynamoDB database in order to store image and blob metadata. This release now includes support for *Redis*.

1.17.4 AWS ECR Sync Support With Temporary Token Authentication

AWS ECR can now be used as the upstream registry to mirror from and the configuration allows for an authentication helper.

1.17.5 Sync Exclude Regex

While mirroring from an upstream registry, it is now possible to exclude images with a regex pattern.

1.17.6 Windows Binaries

Native Windows binaries (both amd64 and arm64) are now distributed with the release.

1.17.7 Bug fixes

The following security issue has been fixed in this release.

`GHSA-c37v-3c8w-crq8/CVE-2025-48374`

1.18

1.18.1 Compatibility with other image schema types

A new configuration setting allows you to store images using the schema Docker Manifest v2 Schema v2. With this setting enabled, zot makes no modifications to the image's manifest or digest. Such modifications would otherwise break the image's signature and attestations.

1.18.2 Bug fixes

The following security issue has been fixed in this release.

`GHSA-c9p4-xwr9-rfhxi/CVE-2025-23208`

1.19

1.19.1 Scale-out cluster

You can build a scale-out cluster (proxy/shard based on repository name). Scale-out cluster is compatible with "sync" feature.

1.19.2 Bug fixes

The following security issue has been fixed in this release.

`GHSA-55r9-5mx9-qq7r/CVE-2024-39897`

1.20

This is a maintenance release with minor bug fixes.

1.21

This is a maintenance release with minor bug fixes.

1.22

1.22.1 CVE Query Enhancements

It is now possible to bisect CVEs (`zli cve diff`) between two image tags/versions in the same repository. Furthermore, a CVE query for a particular image tag can return a detailed description of CVEs.

1.22.2 Documentation for "Immutable Image Tags"

A new article has been added to document how image tags can be made immutable.

1.22.3 Cross-repo tag search in UI

You can now search for a tag across all repos by starting your query as ':' in the UI, which will return all images that have that tag.

1.22.4 Support for ORAS Artifacts removed

OCI distribution spec 1.1.0 has added support "artifacts" which is likely to gain wider adoption. ORAS artifacts are not widely used or supported.

```
:warning:  Support is removed starting from this version.
```

1.23

1.23.1 Support for hot reloading of LDAP credentials file

Since v2.0.0, LDAP credentials have been specified in a separate file. Starting with this version, the file is watched and changes applied without restarting zot.

1.23.2 Bugfixes and performance improvements

Under some configurations, zot consumes significant CPU and memory resources. This has been fixed in this release.

1.24

1.24.1 Updated OCI support

- Support is added for OCI Distribution Spec v1.1.0-rc3 and OCI Image Spec v1.1.0-rc4. The OCI changes are summarized here. These specifications allow arbitrary artifact types and references so that software supply chain use cases can be supported (SBOMs, signatures, etc). Currently, `oras` and `regclient` support this spec.
- For a demonstration of an end-to-end OCI artifacts workflow, see Software Provenance Workflow Using OCI Artifacts.

  ⚠️ Support is deprecated for earlier OCI release candidates.

1.24.2 Built-in UI support

- Using the new zot GUI, you can browse a zot registry for container images and artifacts. The web interface provides the shell commands for downloading an image using popular third-party tools such as docker, podman, and skopeo.

1.24.3 Support for social logins

- Support is added for OpenID authentication with GitHub, Google, and dex.

1.24.4 Group policies for authorization

- When creating authorization policies, you can assign multiple users to a named group. A group-specific authorization policy can then be defined, specifying allowed access and actions for the group.

  ✏️ **Configuration syntax change**: In the previous release, authorization policies were defined directly under the `accessControl` key in the zot configuration file. With the new ability to create authorization groups, it becomes necessary to add a new `repositories` key below `accessControl`. Beginning with zot v2.0.0, the set of authorization policies are now defined under the `repositories` key.

1.24.5 Signature verification

- The validity of an image's signature can be verified by zot. Users can upload public keys or certificates to zot.

1.24.6 LDAP credentials stored separately from configuration

- The LDAP credentials are removed from zot's LDAP configuration and stored in a separate file. See zot's LDAP documentation.

  ⚠️ This LDAP configuration change is incompatible with previous zot releases. When upgrading, you must reconfigure your LDAP credentials if you use LDAP.

1.24.7 Storage deduplication on startup

- Deduplication, a storage space saving feature, now runs or reverts at startup depending on whether the feature is enabled or disabled. You can trigger deduplication by enabling it and then restarting zot.

1.24.8 Retention policies

- To optimize image storage, you can configure tag retention policies to remove images that are no longer needed.

1.24.9 CVE scanning support for image indexes

- The `trivy` backend now supports vulnerability scanning of image indexes. Previously, only images were scanned.

1.24.10 Bookmarks

- In the zot GUI, you can bookmark an image so that it can be easily found later. Bookmarked images appear in search queries when the bookmarked option is enabled.

1.24.11 Ability to delete tags from the UI

1.24.12 Command line search

- The `zli search` command allows smart searching for a repository by its name or for an image by its repo:tag.

1.24.13 Search by digest

- You can perform a global search for a digest (SHA hash) using either the UI or the CLI. This function is useful when an issue is found in a layer that is used by multiple images. In the UI Search box, for example, begin typing `sha256:` followed by a partial or complete digest value to see a dropdown list of images that contain the layer with the digest value.

1.24.14 GraphQL support for search

- A GraphQL backend server within zot's registry search engine provides efficient and enhanced search capabilities. In addition to supporting direct GraphQL queries through the API, zot hosts the GraphQL Playground, which provides an interactive graphical environment for GraphQL queries.

1.24.15 Scheduling of background tasks

- You can adjust the background scheduler based on your deployment requirements for tasks that are handled in the background, such as garbage collection. See Configuring zot.

1.24.16 Performance profiling for troubleshooting

- You can use zot's built-in profiling tools to collect and analyze runtime performance data.

1.24.17 Binaries for FreeBSD

- zot binary images are available for the FreeBSD operating system. Supported architectures are amd64 and arm64.

  ✏️ zot container images for FreeBSD will be available in a future release.

1.25

1.25.1 Remote-only Storage Support

- The two types of state (images and image metadata) can both now be on remote storage so that zot process lifecycle and its storage can be managed and scaled independently.

1.25.2 Digest Collision Detection During Image Deletion

- When two or more image tags point to the same image digest and the image is deleted by digest causes data loss and dangling references. A new behavior-based policy called *detectManifestCollision* was added to prevent this.

August 4, 2026

2. General

2.1 Concepts

2.1.1 What is zot?

👉 **zot** is a production-ready, open-source, vendor-neutral container image registry server based purely on OCI standards.

Two broad trends are changing how we build, distribute, and consume software. The first trend is the increasing adoption of container technologies. The second trend is that software solutions are being composed by combining elements from various sources rather than being built entirely from scratch. The latter trend raises the importance of software provenance and supply chain security. In both trends, zot intends to play an important role by providing a production-ready, open-source, vendor-neutral container image registry server based purely on OCI standards.

What is an OCI image registry?

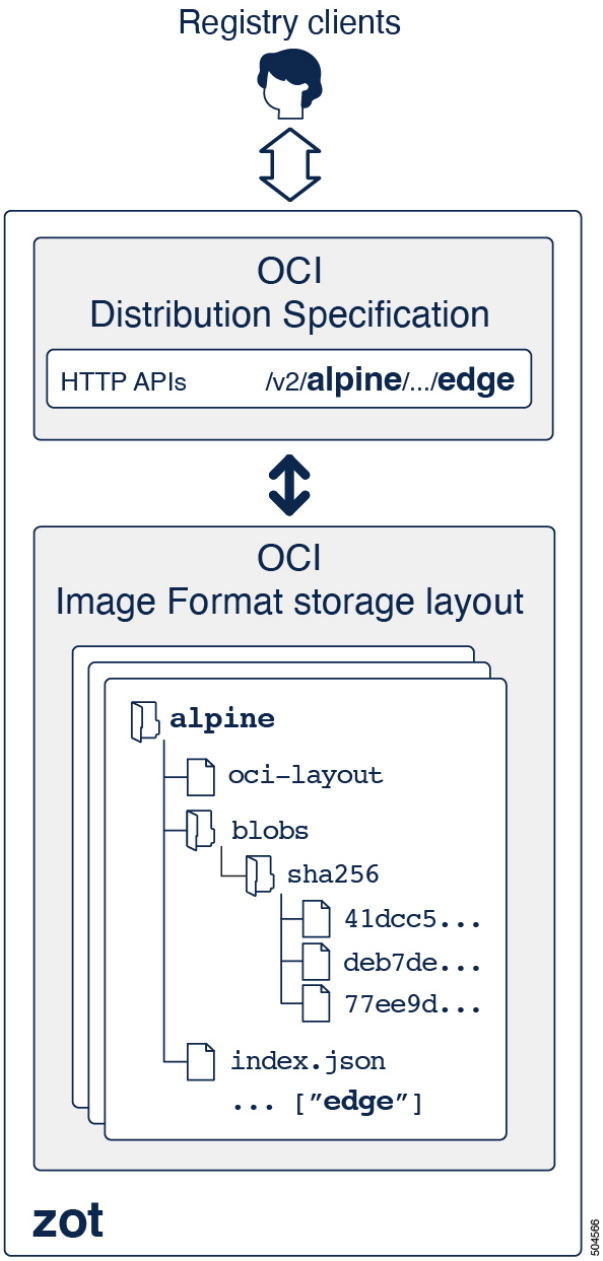
An OCI image registry is a server-based application that allows you to store, manage, and share container images. A developer uploads (pushes) an image to the registry for distribution. Users can then download (pull) the image to run on their systems. The OCI Distribution Specification, published by the Open Container Initiative (OCI), defines a standard API protocol for these and other image registry operations.

An image registry can be a part of your continuous integration and continuous deployment (CI/CD) pipeline when you host zot on your public or private server. In its minimal form, you can also embed a zot registry in a product. In either case, zot provides a secure software supply chain for container images.

2.1.2 Why zot?

👉 zot = OCI Distribution Specification + OCI Image Format

At its heart, zot is a production-ready, vendor-neutral OCI image registry with images stored in the OCI image format and with the OCI distribution specification on-the-wire. zot is built for developers by developers, offering features such as minimal deployment using a single binary image, built-in authentication and authorization, and inline garbage collection and storage deduplication.



Some of the principal advantages of zot are:

- Open source
- OCI standards-only both on-the-wire and on-disk
- Clear separation between core distribution spec and zot-specific extensions
- Software supply chain security, including support for cosign and notation
- Security hardening
- Single binary with many features built-in
- Suitable for deployments in cloud, bare-metal, and embedded devices

zot fully conforms to the OCI Distribution Specification.

The following table lists additional advantages of zot:

Distribution Spec conformance	yes
CNCF project	accepted as a Sandbox Project
License	Apache 2.0
On-premises deployment	yes
OCI conformance*	yes
Single binary image*	yes
Minimal build*	yes
Storage Layout	OCI v1 Image Layout
Authentication	built-in
Authorization	built-in
Garbage collection	inline
Storage deduplication	inline
Cloud storage support	yes
Delete by tag	yes
Vulnerability scanning	built-in
Command line interface (cli)	yes
UI	yes
External contributions	beta available
Image signatures	built-in

✏️ * The **minimal build** feature is the ability to build a minimal Distribution Spec compliant registry in order to reduce library dependencies and the possible attack surface.

May 18, 2023

2.2 Summary of Key Features

- Conforms to OCI distribution spec APIs
- Uses OCI image layout for image storage
- Can serve any OCI image layout as a registry
- Single binary for **all** the features
- Doesn't require *root* privileges
- Clear separation between core dist-spec and zot-specific extensions
- Supports container image signatures - cosign and notation
- Supports helm charts
- Behavior controlled via configuration
- Binaries released for multiple operating systems and architectures
- Supports advanced image queries using *search* extension
- Supports image deletion by tag
- Currently suitable for on-premises deployments (e.g. colocated with Kubernetes)
- Compatible with ecosystem tools such as skopeo and cri-o
- Vulnerability scanning of images
- TLS support Authentication via:
- TLS mutual authentication
- HTTP *Basic* (local *htpasswd* and LDAP)
- HTTP *Bearer* token
- Supports Identity-Based Access Control
- Supports live modifications on the configuration file while zot is running (Authorization configuration only)
- Inline storage optimizations:
- Automatic garbage collection of orphaned blobs
- Layer deduplication using hard links when content is identical
- Data scrubbing
- Serve multiple storage paths (and backends) using a single zot server
- Pull and synchronize from other dist-spec conformant registries
- Supports rate limiting including per HTTP method
- Metrics with Prometheus
- Using a node exporter in case of minimal zot
- Swagger based documentation
- zli: command-line client support
- zb: a benchmarking tool for dist-spec conformant registries
- Released under Apache 2.0 License

September 13, 2023